

Audio Feature Extraction using Signals & Systems

These notes explain the feature-extraction pipeline used to convert raw audio into numerical features that can later be given to machine-learning models.

1. Overall pipeline

Raw audio → Sampling → Discrete-time signal → STFT → Spectrogram → Magnitude/Power spectrum → Spectral features + MFCCs → Feature table → ML models

2. Raw audio and sampling

Sound captured by a microphone can be represented as a continuous-time signal, written as $x(t)$. A computer cannot directly work with a continuous signal, so the signal is sampled at regular time intervals.

$$x[n] = x(nT_s)$$

Here, T_s is the sampling period and F_s is the sampling frequency, with $T_s = 1/F_s$.

For this project, the sampling frequency is $F_s = 22,050$ Hz. This means the audio is represented using 22,050 samples every second.

3. Nyquist-Shannon sampling theorem

The Nyquist-Shannon theorem says that a band-limited signal can be reconstructed without aliasing when the sampling frequency is at least twice the highest frequency present in the signal.

$$F_s \geq 2F_{\max}$$

Therefore, the Nyquist frequency for a 22,050 Hz sampling rate is:

$$F_{\text{Nyquist}} = F_s/2 = 22,050/2 = 11,025 \text{ Hz}$$

So frequencies up to about 11,025 Hz can be represented without aliasing, assuming the signal is appropriately band-limited and anti-alias filtering is used before sampling.

4. Why a normal Fourier Transform is not enough

A normal Fourier Transform tells us which frequencies are present in the complete signal, but it does not directly tell us when those frequencies occur.

This is a problem for music because music is **non-stationary**. Its frequency content changes with time. For example, a song may contain a piano section, followed by guitar and then drums. A single Fourier Transform mixes the frequency information from all of these sections together.

To keep both time and frequency information, the Short-Time Fourier Transform (STFT) is used.

5. Short-Time Fourier Transform (STFT)

The STFT divides the audio into short, usually overlapping frames. A window function is applied to each frame and a Fourier transform is then calculated for that frame.

$$\text{Windowed frame} = x[n]w[n]$$

The window is then shifted along the signal and the process is repeated. Common window functions include the Hann and Hamming windows.

The main idea is simple: instead of finding the frequency content of the whole recording at once, we find the frequency content of many small sections.

6. DFT and spectrogram

For each short frame, the Discrete Fourier Transform (DFT) converts the signal from the time domain to the frequency domain.

$$X[k] = \sum x[n] e^{-j2\pi kn/N}$$

The result is a frequency spectrum for each frame. Putting these spectra together gives a **spectrogram**.

A spectrogram shows how the strength of different frequencies changes with time. In simple terms, it contains time, frequency and magnitude information.

7. Magnitude and power spectrum

The DFT produces complex-valued values. The magnitude of a frequency component is written as $|X[k]|$ and indicates how strong that component is.

$$|X[k]| \rightarrow \text{magnitude spectrum}$$

$$P[k] = |X[k]|^2 \rightarrow \text{power spectrum}$$

These representations are useful for calculating audio descriptors such as the spectral centroid and for the later MFCC calculation.

8. Spectral centroid

The spectral centroid can be thought of as the centre of mass of the frequency spectrum. It is a weighted average of the frequencies, where the magnitude of each frequency acts as the weight.

$$SC = \sum f_k |X[k]| / \sum |X[k]|$$

A higher spectral centroid generally means that more energy is concentrated toward higher frequencies. This is often associated with a brighter or sharper sound. A lower centroid is often associated with a darker or more mellow sound.

The centroid does not directly measure human perception of brightness; it is a numerical feature that often correlates with it.

9. Timbre

Timbre is the quality or character of a sound that helps us distinguish between different instruments even when they play the same note at the same pitch.

For example, a piano and a guitar can both play 440 Hz, but they still sound different because their harmonic content and spectral envelopes are different.

10. MFCCs

MFCC stands for **Mel-Frequency Cepstral Coefficient**. MFCCs are widely used in speech and audio processing because they give a compact representation of important spectral characteristics related to human hearing and timbre.

Basic MFCC flow: Power spectrum $\rightarrow$ Mel filter bank $\rightarrow$ Log $\rightarrow$ DCT $\rightarrow$ MFCCs

11. Mel scale and Mel filter bank

Human perception of frequency is not linear. The Mel scale is used to represent frequency in a way that roughly follows how humans perceive pitch and frequency differences.

A Mel filter bank contains overlapping filters that group nearby frequency components. The energy from each filter is calculated, producing a smaller and more perceptually meaningful set of values than the original frequency spectrum.

12. Logarithm in MFCC

After the Mel filter-bank energies are obtained, a logarithm is applied. This compresses the dynamic range and is also more consistent with the approximately logarithmic way humans perceive sound intensity.

Mel energies $\rightarrow \log(\text{Mel energies})$

13. DCT and decorrelation

The Discrete Cosine Transform (DCT) is applied to the log Mel energies. The DCT concentrates important information into a smaller number of coefficients and reduces redundancy between neighbouring filter-bank values.

The resulting coefficients are the MFCCs. Usually, only the first several coefficients are retained as features.

14. Final feature table

After calculating descriptors such as spectral centroid and MFCCs, the values can be arranged into a tabular dataset. Each row can represent an audio sample or recording, while the columns contain numerical features.

Sample	Spectral Centroid	MFCC 1	MFCC 2	MFCC 3	Class
1	0.42	-12.4	3.1	-1.7	Class A
2	0.68	-8.2	1.9	-0.8	Class B
3	0.51	-10.7	2.4	-1.1	Class A

15. One-minute viva explanation

I start with continuous-time audio $x(t)$ and sample it at 22,050 Hz to obtain a discrete-time signal $x[n]$. The Nyquist frequency is therefore 11,025 Hz. Since music is non-stationary, I use STFT rather than a single Fourier Transform. The signal is divided into short overlapping frames, windowed, and transformed using the DFT. The spectra from these frames form a spectrogram. From the magnitude or power spectrum I calculate spectral descriptors such as the spectral centroid. To capture timbral information, I calculate MFCCs by mapping the power spectrum through Mel filter banks, taking the logarithm, and applying the DCT. These numerical descriptors form the final feature table used by the machine-learning models.

16. Quick revision table

Concept	What it does
Sampling	Converts continuous-time audio into discrete samples
Nyquist theorem	Sets the minimum sampling rate needed to avoid aliasing

DFT	Converts a frame from time domain to frequency domain
STFT	Shows how frequency content changes with time
Spectrogram	Visual representation of STFT magnitude/power
Spectral centroid	Weighted centre of the frequency spectrum
Timbre	Character/quality that distinguishes sounds
Mel scale	Frequency scale designed around human perception
MFCC	Compact representation of important spectral/timbral information
DCT	Compresses and decorrelates log Mel features